

Lab 3: Combinational Logic Circuit

In this lab, it was asked to set up a combinational logic circuit on the breadboard.

Components that are used in this lab:

- A 4-bit binary counter (74HC163 Quad)
- A 2 input AND Gate (74HC08 Quad)
- A input OR Gate (74HC32 Quad)
- A resistor
- A LED light

According to the datasheet for the 74HC163 counter, the 4-bit binary counter is placed on the breadboard. The positive side of the breadboard is linked to Vcc (This procedure is done for all the gates used in this circuit.) The 4-bit binary counter's Q0, Q1, Q2, and Q3 inputs are utilized to watch and acquire a testbench. A two-input AND Gate is positioned after the counter.

The equation of the circuit designed in this lab can be written as $XY + Z$ (Where X is input-1, Y is input-2, Z is input-3)

INPUT 1	INPUT 2	INPUT 3	OUTPUT
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Table 1: Truth Table of the Circuit

Q0 on the 74HC163 Quad is connected to 1 on the 74HC08 Quad through jumper wires, while Q1 on the 74HC163 Quad is connected to 2 on the 74HC08 Quad via jumper cables. On the 74HC08 Quad, 1 and 2 are the inputs, and 3 is the output produced through the AND Gate.

A two-input OR Gate is placed after the AND Gate. Using jumper cables, Q2 on 74HC163 Quad is connected to 1 on 74HC32 Quad and the output produced from the AND Gate (3 on 74HC08 Quad) is connected to 2 on 74HC32 Quad. The circuit's inputs are 1 and 2, and the desired output is 3 on the 74HC32 Quad. (Each gate in this circuit has a number 7 connected to ground.) Finally, a resistor and an LED light are connected to physically view the output. The output wave is also observed using an oscilloscope.

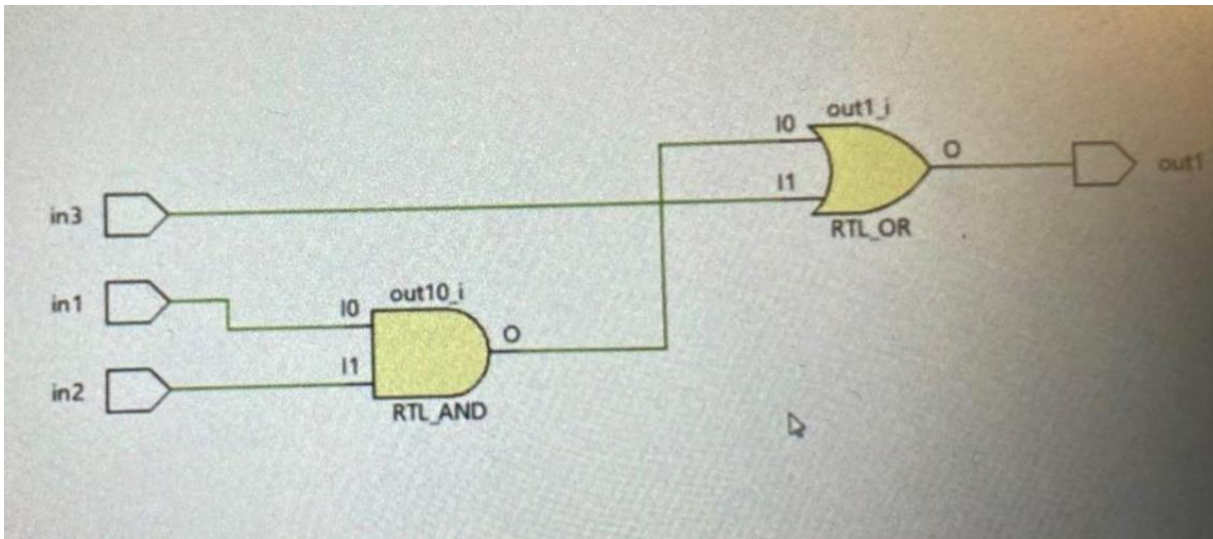


Figure 1: Schematic of the circuit design

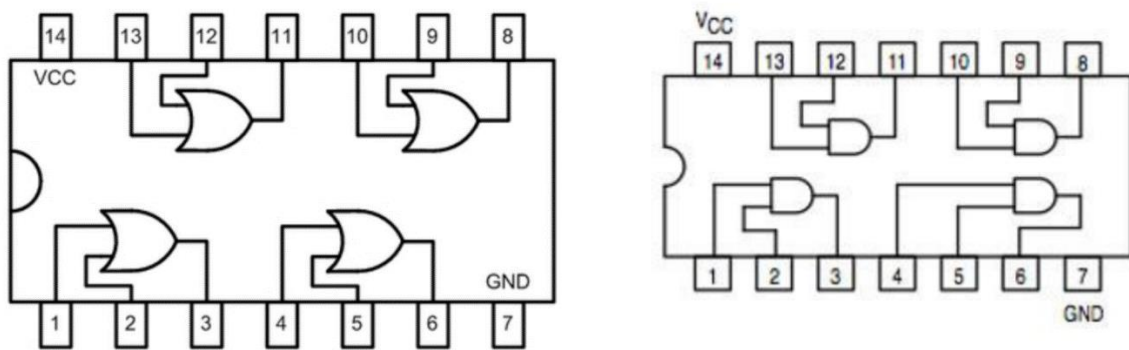


Figure 2-3: OR Gate (74HC32 Quad), AND Gate (74HC08 Quad)

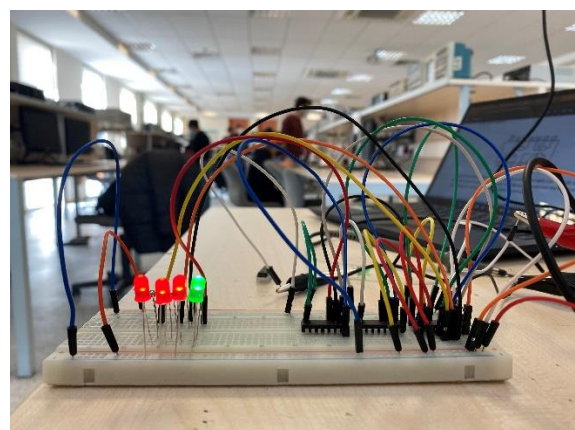
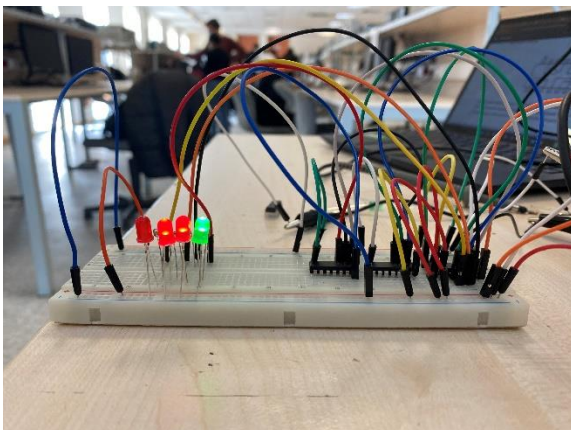


Figure 4-5: $x=y=out=1, z=0$; $x=y=z=out=1$

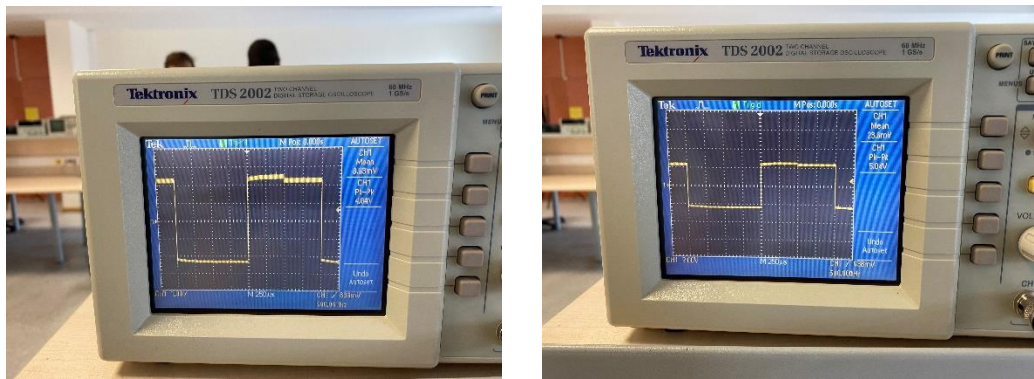


Figure 6-7: Waveforms of Q0 and Q1 of A 4-bit binary counter (74HC163 Quad)